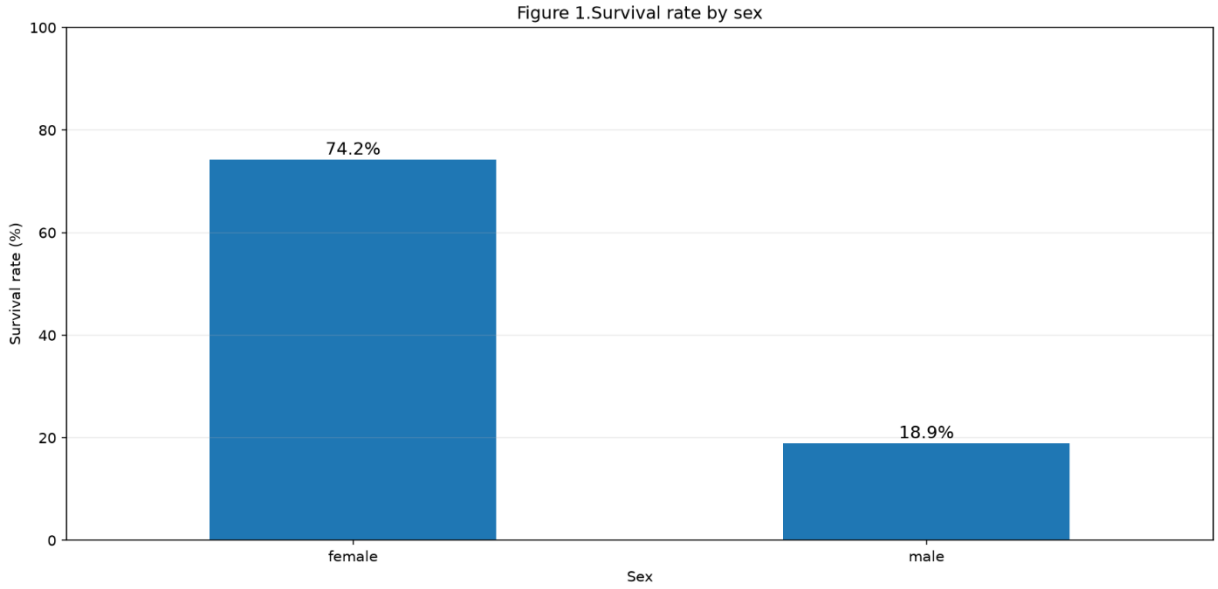
Titanic Passenger Survival Analysis

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1. Introduction and Methodology:

In this report, the authors explore the relationship between passenger characteristics and their chances of surviving the Titanic shipwreck in 1912. It draws on Titanic Training data from Kaggle (891 passengers), focusing on gender, passenger class, and age. “Survived” is Passengers have been coded as 1 for living and 0 for not living. 38.4% of the 342 passengers survived. These results indicate trends in the sample and do not suggest that any one trait caused a particular passenger’s outcome.

2. Differences in survival by sex:



There is a difference between the two groups in Figure 1. 233 out of 314 people survived, or 74.2 percent. 109 people survived out of 577 passengers, a survival rate of 18.9 percent. The survival rate for passengers is about 3.9 times higher than the survival rate for male passengers. This strong link in the data matches the idea that women were given priority during evacuation. Sex may also relate to passenger class – where a person was on the ship and how far a person could access a lifeboat. Thus, the chart does not present a distinction between the effects of sex and those factors.

3. Survival by passenger class:

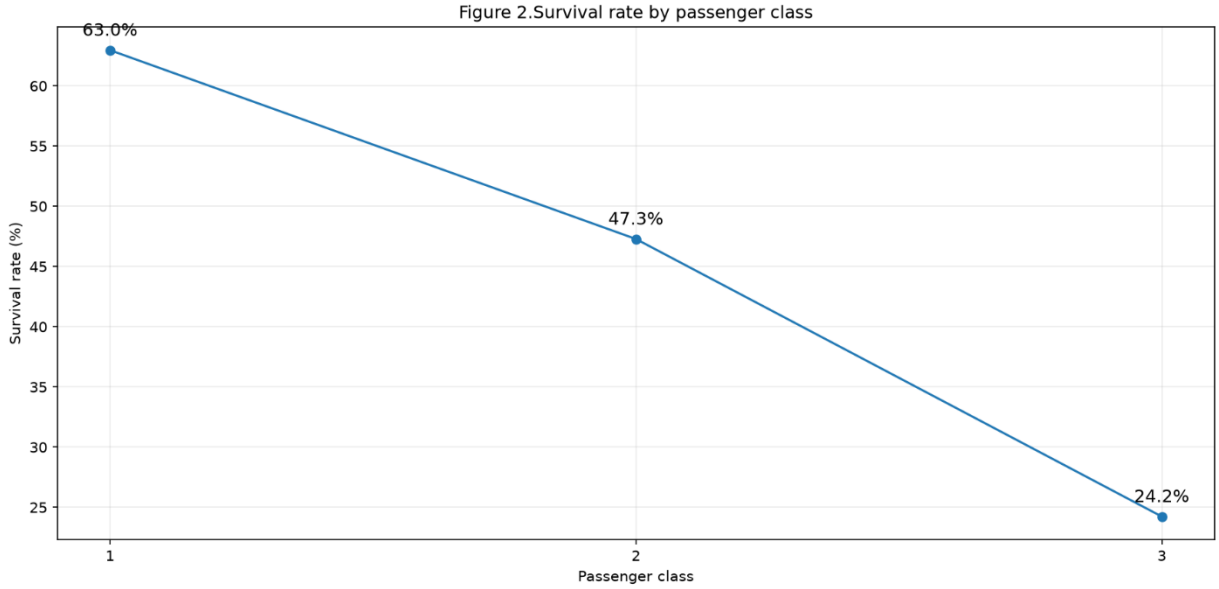
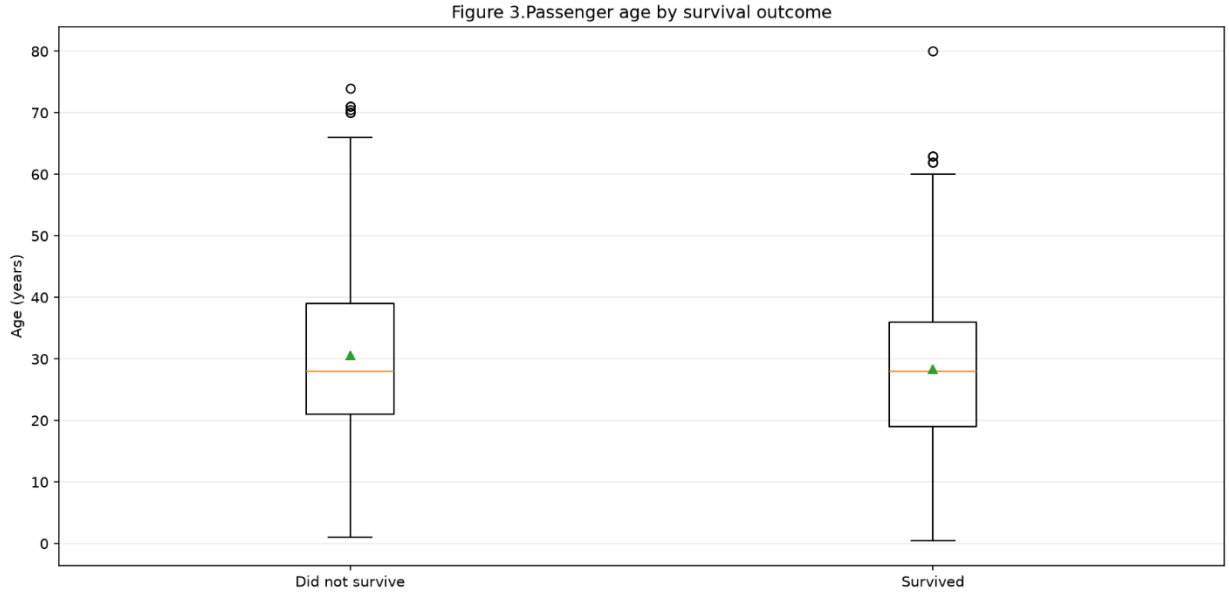


Figure 2 indicates the overall downward passenger class trend. The percentage of survival in class was 63.0 %. Survival was 63.0% in class again. The number of survivors in class was 136 out of 216. There was a 47.3% survival rate in class. 87 out of 184 survived in class. The survival rate was 24.2% in class. The number of survivors in class was 119 out of 491. The class rate was 1/2.6 of the first class rate. The first class rate was 2 ½ times the class rate once more. The pattern indicates that there were some links between social and economic

status with the chances of survival. The pattern indicated that social and economic status were correlated with again being able to survive. The results could be due to variations in the location of cabins near exits and in access to the evacuation area. It might be due to a variation in the proximity of the cabin location to exits and access for evacuation again. The line connecting the ordered categories is drawn to help identify the trend. The line connecting ordered categories is drawn to help see the trend. A line connecting ordered categories is not to be confused as a numerical relationship between the class numbers. Again, there should be no numeric relationship between the line that connects ordered categories and the class numbers.

4. Age distribution and survival rate:

Figure 3.Passenger age by survival outcome



The table in figure 3 displays the ages of passengers who survived and those who perished. There were 424 passengers whose ages were recorded—among them were those who did not survive. These non-survivors were an average of 30.6 years old. The average age was 28. The number of people who survived the flight was 290. Their average age was 28.3 years. The median age was also 28. The medians are equal. The boxes have a lot of overlap. This means that it is not clearly apparent that the two groups are separated by age. On average, the survivors were slightly younger. This difference should be looked, at carefully. For 177 passengers there was no age information. This is the box plot of the 714 records with known age. No missing ages were. Made an educated guess for this comparison.

5. Data preparation and critical evaluation:

I viewed the data set. Found 177 missing age values 687 missing cabin values and 2 missing embarkation port values. In the box plot analysis, I left age as missing of trying to fill in values so that the original age distribution would stay the same. There are some missing cabin data – 891 missing cabin data. Because of that cabin was not used as a factor in the analysis. I worked out survival percentages, for each group, which would be easier to make comparisons between groups of differing sizes rather than the raw numbers. An issue is that these comparisons are not adjusted for sex, class, age, family situation and deck location may be correlated. The data in the training set is of an event rather than from an experiment, which means that it cannot demonstrate cause and effect or be extrapolated to other shipwrecks.

6. Conclusion:

There was an overall survival of 38.4% in this data set. The sex difference was most striking (74.2% of the female passengers survived compared to 18.9% of the male passengers) and the difference by class of passenger was next (63.0% of the first class passengers survived compared to 24.2% of the third class passengers). The available-age sample has a slightly lower mean age among survivors, which does not affect age distributions. The gatherings indicate that the chances of surviving did not seem to be equal for passengers in these groups.

7.Reference:

Kaggle: Titanic - Machine Learning from Disaster.

Available at: https://www.kaggle.com/competitions/titanic/data?select=train.csv

Github : https://github.com/adsmanikanta2000-bit/Titanic-Passenger-Survival-Analysis.git\